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 Studierichting: Informatica
 Oefeningenreeks 5A nr. 5.2

$$\begin{aligned}
 BI \int_{-2}^2 (4x^2 - 5x + 2) dx &\Rightarrow OBI = \int (4x^2 - 5x + 2) dx \\
 &= 4 \int x^2 dx - 5 \int x dx + 2 \int dx = \frac{4x^3}{3} + c - \frac{5x^2}{2} + c + 2x + c = \frac{4}{3}x^3 - \frac{5}{2}x^2 + 2x + c \\
 BI &= \left[\frac{4}{3}x^3 - \frac{5}{2}x^2 + 2x + c \right]_{-2}^2 = \frac{4}{3}(2)^3 - \frac{5}{2}(2)^2 + 2 \cdot 2 - c - \left(\frac{4}{3}(-2)^3 + \right. \\
 &\left. \frac{5}{2}(-2)^2 - 2 \cdot (-2) + c \right) = \frac{32}{3} - \frac{20}{2} + 4 + \frac{32}{3} + \frac{20}{2} - 4 = \frac{64}{3} + 8 = \frac{88}{3}
 \end{aligned}$$

References